

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Krisha Meenatchi M (192511338) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Krisha Meenatchi M

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

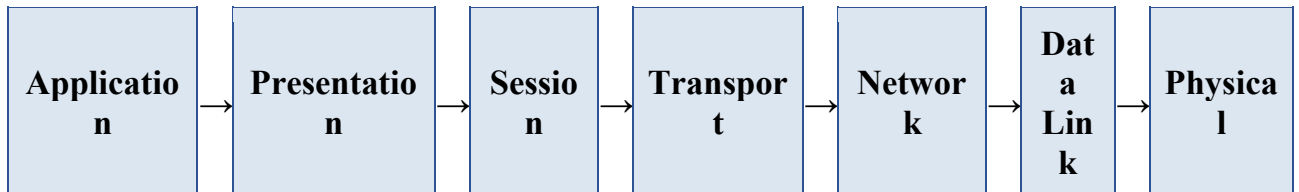
RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Question 1

The OSI (Open Systems Interconnection) model organizes network communication into seven layers, each performing a distinct function while cooperating with the layers directly above and below it. The partially completed chain below has two gaps to fill.

Concept Map



Concept Identification

The two missing layers are the Session Layer (positioned between Presentation and Transport) and the Data Link Layer (positioned between Network and Physical). Together with the five given layers, these complete the full seven-layer OSI stack: Application, Presentation, Session, Transport, Network, Data Link, and Physical.

Linking & Relationships

- Session Layer: establishes, manages, and terminates the dialogue (session) between two communicating applications. It handles synchronization points and checkpoints so that if a connection is interrupted, communication can resume from the last checkpoint instead of restarting from zero.
- Data Link Layer: takes packets handed down from the Network layer and frames them with a header/trailer containing MAC addresses and a frame-check-sequence (CRC), enabling node-to-node delivery across a single physical link and detection of transmission errors.

Application & Analysis — Why these layers are essential for reliability

Without the Session layer, applications would have no mechanism to keep a logical conversation intact across multiple exchanges; a dropped connection would force a complete restart, wasting bandwidth and increasing the chance of data loss during long transfers (e.g., file uploads, video calls). The Session layer's checkpointing directly supports reliability by allowing graceful recovery.

Without the Data Link layer, the Network layer's packets would be handed directly to the Physical layer as raw bits with no addressing for the local segment and no error-detection mechanism. Electrical noise, collisions, or attenuation on the wire would corrupt data silently. The Data Link layer's framing and CRC checks catch these errors at the very first hop, preventing corrupted frames from propagating further up the stack and triggering retransmission at the earliest possible point.

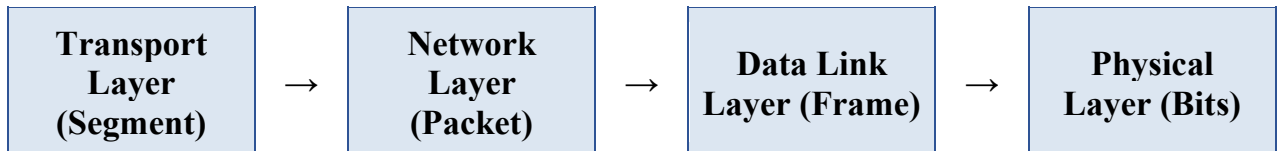
Explanation & Clarity — Summary

Each OSI layer adds a specific safeguard: Session preserves the continuity of the conversation, while Data Link protects the integrity of every individual hop. Together with the other five layers, they ensure that data not only reaches its destination but arrives complete, in order, and without silent corruption — which is the essence of 'reliable data transmission' referenced in CO1.

Question 2

As data moves down the OSI stack for transmission, each layer wraps the data from the layer above with its own control information — a process called encapsulation. The resulting unit at each layer has a specific name.

Concept Map



Concept Identification

- Transport Layer → Segment (TCP) / Datagram (UDP) — adds source & destination port numbers, sequence numbers.
- Network Layer → Packet — adds source & destination IP addresses for end-to-end logical addressing.
- Data Link Layer → Frame — adds source & destination MAC addresses plus a trailer (FCS/CRC) for local delivery and error detection.
- Physical Layer → Bits — converts the frame into an electrical, optical, or radio signal for transmission over the medium.

Concept Mapping Structure — Encapsulation Flow

Data → Segment → Packet → Frame → Bits. Each arrow represents one layer adding its own header (and, at the Data Link layer, a trailer as well), progressively wrapping the original application data like nested envelopes.

Application & Analysis

This transformation of data units is what allows heterogeneous networks to interoperate. The Transport layer's segment numbering allows the receiving host to reorder and reassemble data even if packets arrive out of sequence. The Network layer's packet, carrying IP addresses, allows routers to make forwarding decisions across multiple networks without needing to understand the application data inside. The Data Link layer's frame, carrying MAC addresses, allows switches to deliver the packet correctly within a single local network segment. Finally, the Physical layer's bit stream is the only truly transmittable form of the data — everything above it is only meaningful after correct decapsulation at the receiver.

At the receiving end, the exact reverse process (decapsulation) strips each header in order — Bits → Frame → Packet → Segment → Data — restoring the original message. If any layer's control information is corrupted or missing, decapsulation fails at that specific layer, which is why understanding the data-unit-to-layer mapping is essential for both correct transmission and effective network troubleshooting.

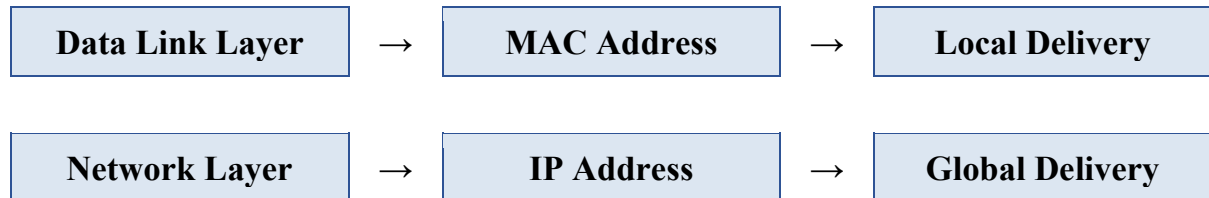
Explanation & Clarity — Summary

The progressive renaming of the data unit (Segment → Packet → Frame → Bits) is not arbitrary terminology; it reflects a genuine structural transformation, with each layer adding exactly the information it needs to perform its job, and no more. This separation of concerns is what makes the OSI model modular and scalable.

Question 3

The given concept map contrasts two parallel delivery mechanisms — one operating at the Data Link layer using physical addressing, and the other at the Network layer using logical addressing.

Concept Map



Concept Identification

- **MAC Address:** a 48-bit hardware address, unique to each network interface card, used only within a single physical/local network (LAN).
- **IP Address:** a logical address (IPv4/IPv6) assigned to a device, used for identifying hosts across interconnected networks (the Internet).

Linking & Relationships

The Data Link layer uses the MAC address to move a frame between two directly connected devices — for example, from a laptop to the nearest router, or between two switch ports. This is 'local delivery' because a MAC address has no meaning outside the local network segment; routers do not forward frames based on MAC addresses.

The Network layer uses the IP address to identify the ultimate source and destination hosts, regardless of how many intermediate networks the data must cross. This is 'global delivery' because IP addressing is hierarchical and routable — routers use it to make forwarding decisions hop by hop until the packet reaches the destination network.

Application & Analysis — How they ensure complete delivery

These two addressing schemes work together, not in isolation. When a host sends data to a remote server, the IP address never changes throughout the journey — it represents the true end-to-end destination. However, the MAC address changes at every hop: each router replaces the frame's source and destination MAC addresses with those of the next local link (using ARP to resolve the next hop's MAC address), while leaving the IP header untouched.

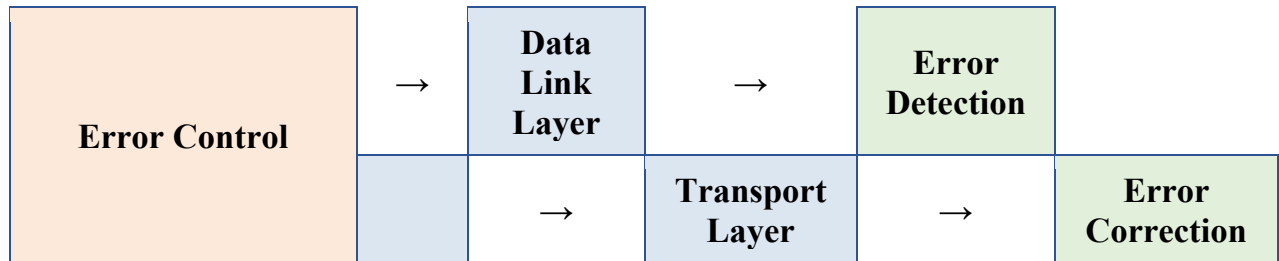
This layered division of labour is what allows a single packet to traverse many different physical networks (Ethernet, Wi-Fi, fiber backbones) that each have their own local addressing, while still arriving at exactly the correct final host. Without MAC addressing, devices on a shared local segment could not distinguish frames intended for them; without IP addressing, there would be no consistent identifier to route data across the boundaries between local networks.

Explanation & Clarity — Summary

In short: IP address = 'where in the world', MAC address = 'who is next door.' Global delivery is achieved through a sequence of local deliveries, each governed by a freshly resolved MAC address, while the IP address preserved unchanged, guarantees the data reaches the correct final destination.

Error control is not the responsibility of a single OSI layer; it is distributed so that lightweight checks happen close to the physical medium, while more thorough recovery happens closer to the application.

Concept Map



Concept Identification

- Data Link Layer — Error Detection: uses a Frame Check Sequence (CRC) computed over each frame; the receiver recomputes the CRC and discards the frame if it does not match, but does not itself request retransmission in most modern LAN designs.
- Transport Layer — Error Correction: uses sequence numbers, acknowledgements (ACKs), timers, and retransmission (as in TCP) to guarantee that lost or corrupted segments are detected end-to-end and re-sent until confirmed delivered.

Concept Mapping Structure & Relationships

The map branches from a single parent concept, 'Error Control,' into two child processes at two different layers. This hierarchical structure reflects that error handling is layered defense-in-depth: the Data Link layer's detection is fast and local (per single hop), while the Transport layer's correction is slower but comprehensive (per end-to-end connection).

Application & Analysis

Consider a large file transferred over multiple hops. At every hop, the Data Link layer's CRC check catches bit errors introduced by electrical noise on that specific link and silently drops the corrupted frame — this prevents obviously damaged data from wasting bandwidth on subsequent hops. However, a dropped frame at one hop is invisible to the original sender; the Data Link layer alone cannot guarantee that the complete file eventually arrives.

This is where the Transport layer's error correction becomes essential: because it operates end-to-end between the original sender and final receiver, it can detect a missing segment (via gaps in sequence numbers or a missing ACK before a timer expires) no matter which intermediate hop dropped it, and can trigger a fresh retransmission of just that segment. The interaction between the two layers therefore combines fast, local filtering of bad data with a slower, global guarantee that no data is permanently lost — a classic example of layered reliability that no single layer could achieve alone.

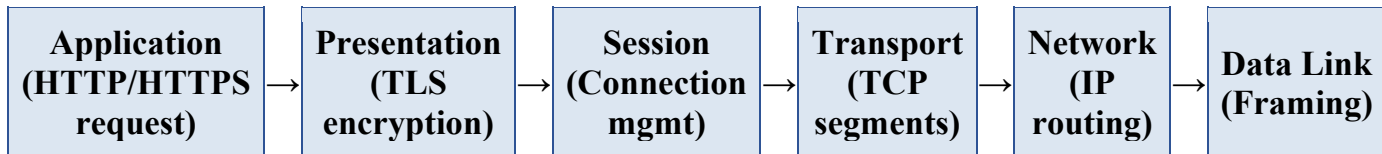
Explanation & Clarity — Summary

Data Link error detection acts as a first line of defense that keeps obviously corrupted frames from propagating; Transport layer error correction acts as the final safety net that guarantees complete, correct delivery of the whole message, even when individual frames are lost anywhere along the path.

Question 5

A simple act such as loading a web page actually depends on all seven OSI layers cooperating correctly, each contributing one necessary function.

Concept Map — Web Browsing across OSI Layers



Concept Identification

- Application: the browser generates an HTTP/HTTPS GET request for the requested URL.
- Presentation: TLS encrypts/decrypts and the content is formatted (e.g., compression, character encoding) for HTTPS.
- Session: manages the TLS session and keeps the logical dialogue open across multiple requests to the same server.
- Transport: TCP breaks the request into segments, establishes the three-way handshake, and ensures ordered, reliable delivery.
- Network: IP addressing and routing get the packets from the client's network to the web server's network.
- Data Link: frames the packets with MAC addressing for each physical hop (home router, ISP switches, etc.).
- Physical: converts frames into electrical, optical, or radio signals across cables, fiber, or Wi-Fi.

Application & Analysis — Impact of Layer Failure

Because the layers form a dependency chain, a failure at any single layer breaks the entire communication process, though the visible symptom differs by layer:

- Physical failure (unplugged cable, Wi-Fi out of range): no signal at all — the browser simply times out with no data exchanged.
- Data Link failure (wrong MAC resolution, switch misconfiguration): frames never reach the correct local device, so packets are silently dropped before even leaving the local network.
- Network failure (routing misconfiguration, wrong IP/gateway): packets may leave the local network but never find a path to the destination server, producing a 'destination unreachable' type error.
- Transport failure (blocked port, firewall dropping TCP handshake): the connection cannot be established even though routing succeeds, so the browser reports it cannot connect to the server.
- Session/Presentation failure (TLS handshake or certificate error): the connection reaches the server but data cannot be securely interpreted, producing a security-warning or blank page.

- Application failure (malformed HTTP request, server returns an error code): the page fails to render even though every lower layer worked perfectly, e.g., an HTTP 404 or 500 response.

Explanation & Clarity — Summary

Web browsing is a useful illustration of the OSI model precisely because a single click involves every layer in sequence, both on the way out and on the way back. A malfunction anywhere in that sequence halts the entire process, which is why understanding this concept map is directly useful for diagnosing 'why won't this website load' in practice.

Question 6

The concept map below represents a network diagnostic snapshot, where the first two layers are functioning correctly (✓) and the remaining three layers are failing (✗).

Concept Map — Diagnostic Snapshot



Concept Identification

The pattern of ticks and crosses is diagnostic in itself: the two lowest layers (Physical and Data Link) work correctly, while the Network layer is the first point of failure, with Transport and Application also failing as a direct consequence.

Linking & Relationships — Root Cause Analysis

Because Physical and Data Link both succeed, the cabling, network interface card, signal, and local addressing (MAC resolution, switching within the LAN) are confirmed to be working — the device can successfully exchange frames with its directly connected neighbours, such as its default gateway.

The failure begins precisely at the Network layer. Likely causes include: an incorrect IP address or subnet mask, a missing or wrong default gateway, a routing table misconfiguration, or the destination network being genuinely unreachable. Because the Network layer is where global addressing and routing happen, its failure means packets cannot be forwarded beyond the local segment.

Application & Analysis — Cascading Effect

Transport and Application layer failures here are not independent problems; they are downstream symptoms of the same root cause. TCP (Transport) cannot complete its handshake or deliver segments if IP packets never leave the local network, so it correctly reports failure. Likewise, the Application layer (e.g., a browser or any client software) cannot function because it depends entirely on a working Transport connection to exchange any data at all.

This is precisely why concept maps are valuable troubleshooting tools: by visually separating each OSI layer's status, a network engineer can immediately see that effort should be focused on the Network layer (checking IP configuration, default gateway, and routing) rather than wasting time investigating the Application or Transport layers, whose failures are merely consequences, not causes.

Explanation & Clarity — Summary

The concept map isolates the fault to a single layer by showing exactly where the chain of ✓ marks breaks. Troubleshooting should always begin at the lowest failing layer, since every layer above it depends on that layer's function; fixing the Network layer issue here should restore the Transport and Application layers automatically, without any changes being needed at those higher layers.